MCS24 TEAM

FACEGUARD

PRIVACY-PRESERVING AI FOR DETECTING DEEPFAKE PROFILE PHOTOS

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TABLE OF CONTENTS

Project Introduction

Project Proposal

Project Management

Project Conclusion

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INTRODUCTION

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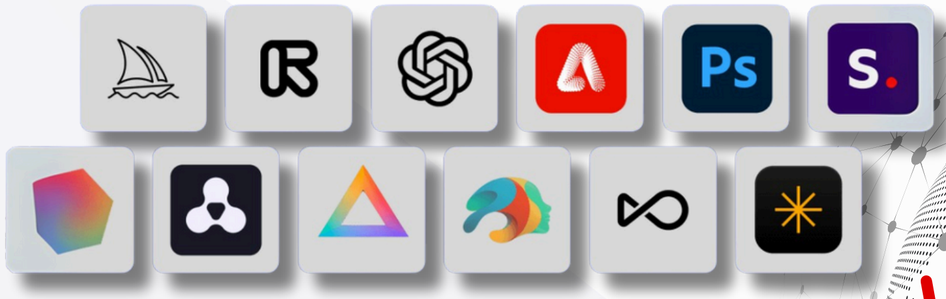
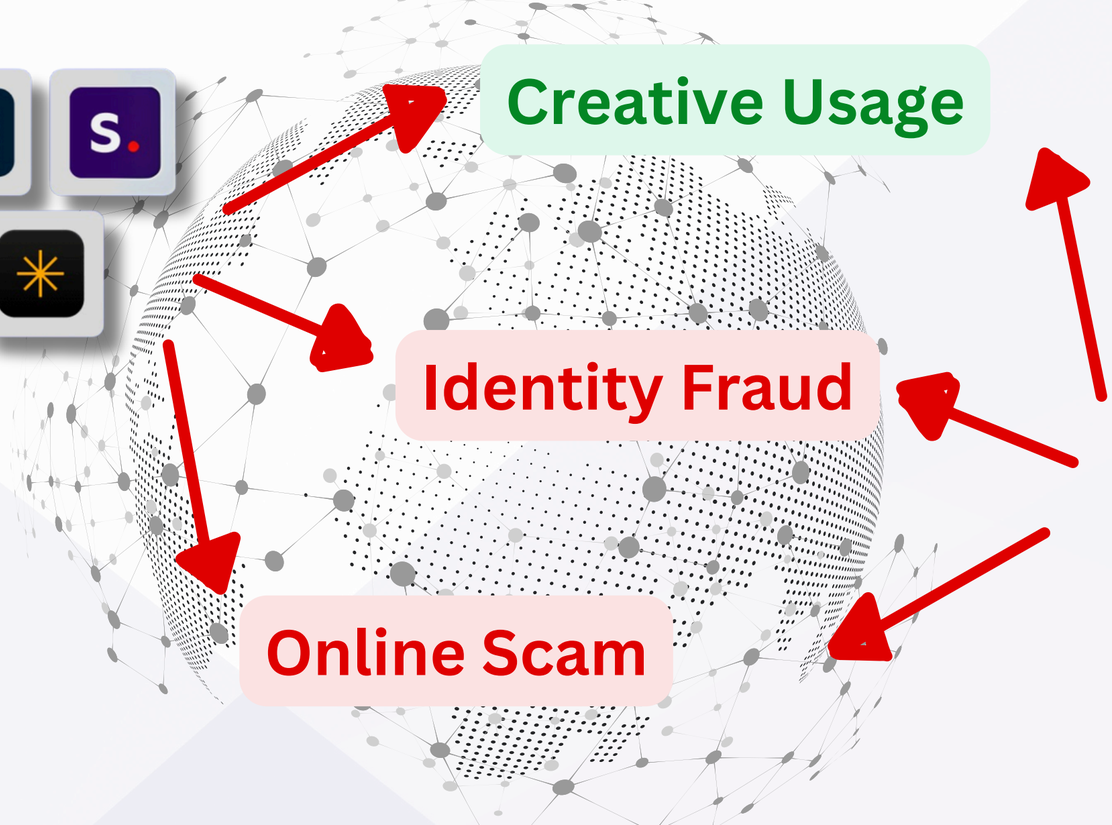
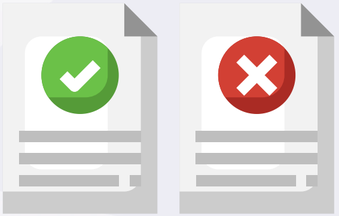
INTRODUCTION

Rapid Advancement of Gen AI (GANs & Diffusion Model)

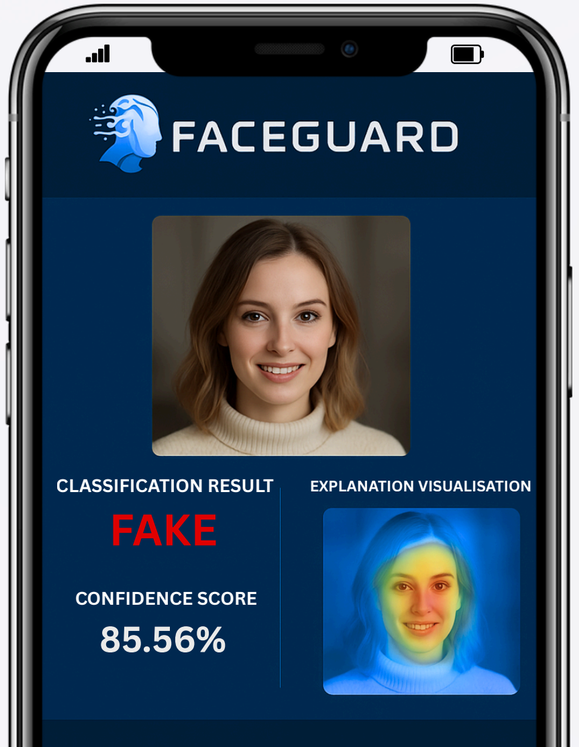


Widespread of AI-generated Faces in Online World





AI-powered Detector



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PROJECT GOALS

FACEGUARD

01. Strengthen trust and safety in online interactions

02. Protect individuals from fraudulent profiles

03. Promote responsible AI practices

04. Foundation for future application in fraud prevention

TARGET AUDIENCE

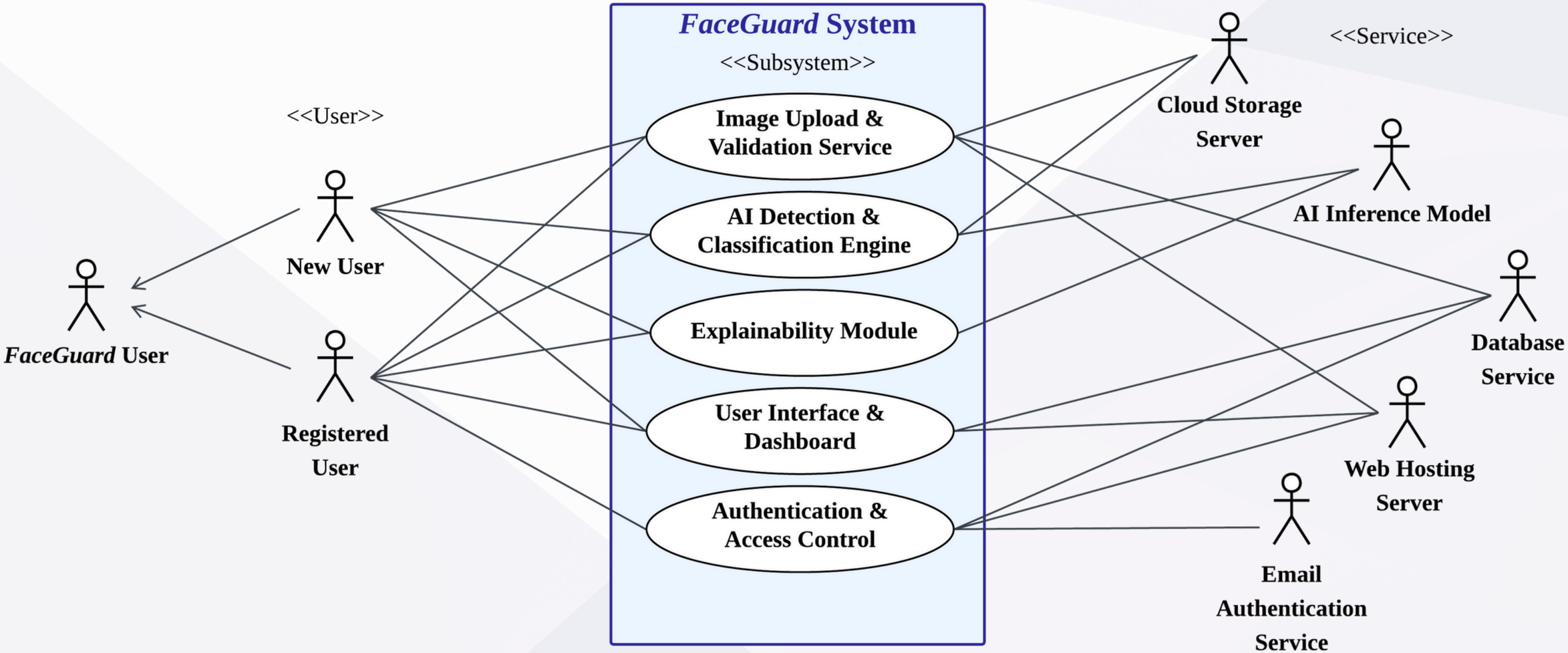
Target Users

Online Users
(Teenagers / Young adults)

Stakeholders

- **Social Media Platforms**
- **Dating Apps**
  (Potential integration)

PROJECT USE CASES



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PROPOSAL

PROJECT REQUIREMENTS

Real vs AI Image Classification

Model Training | FR | High

- *FaceGuard*'s **core functionality**
- Verify **image authenticity**

Secure Image Upload Interface

Security | FR | High

- **Image deleted** after processing

User-Friendly Dashboard Design

Usability | FR | High

- **Drag-and-drop** upload interface
- Clear **visual** explanations
- **Get started** guidebook

Confidence Scoring & Explainability

Web Application | FR | High

- Classification result with a **confidence score** and an **explainability visualisation**

Fairness & Inclusivity in Model Performance

Model Evaluation | FR | Low

- **Bias-mitigation** techniques to maintain equitable outcomes
- **Optional requirement** to fulfill

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PROJECT OVERVIEW

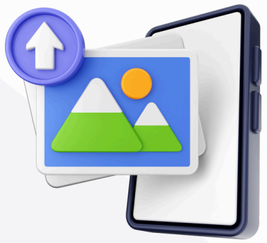


Image Upload

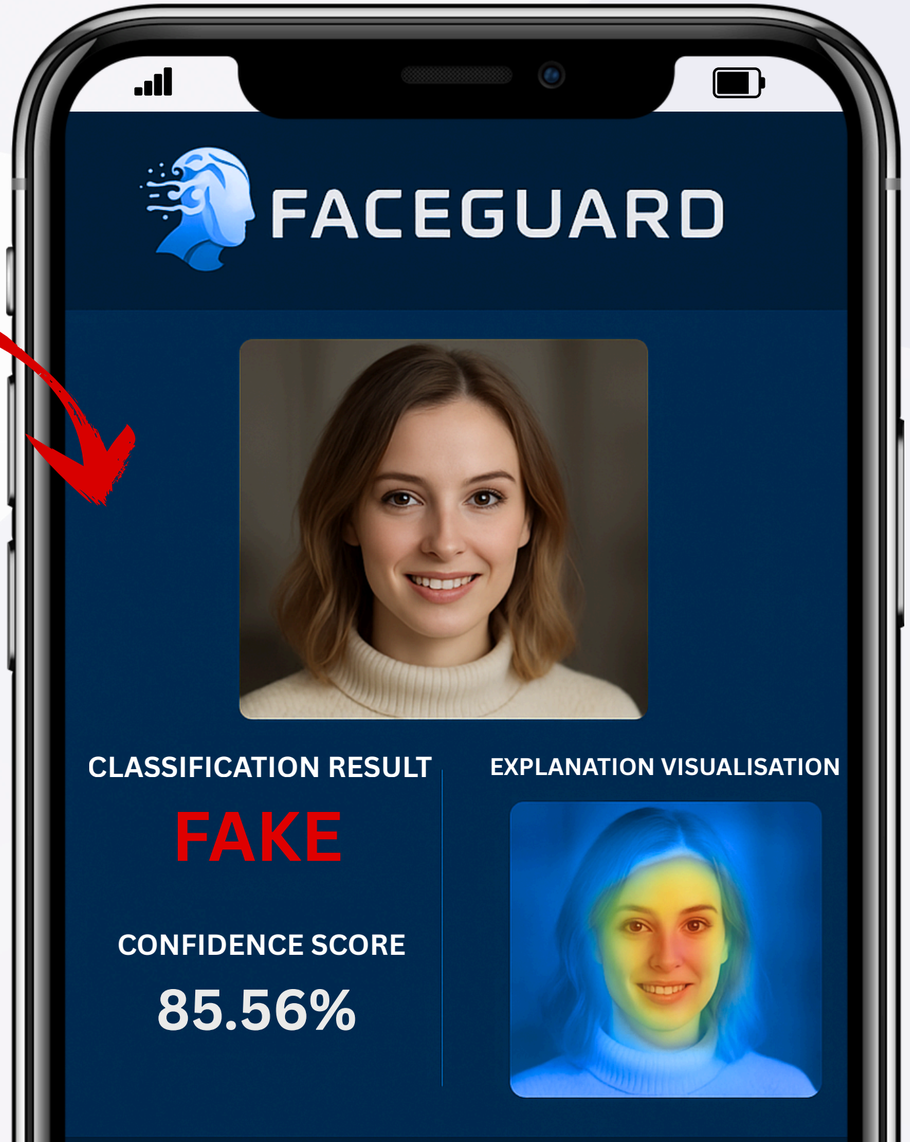
Classifier Analysis

REAL OR FAKE?

Secure Processing

NO STORAGE!!!

DEEPFAKE DETECTED!!!



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PROJECT PROTOTYPE

- Home Page
- Start Guide
- Sign In

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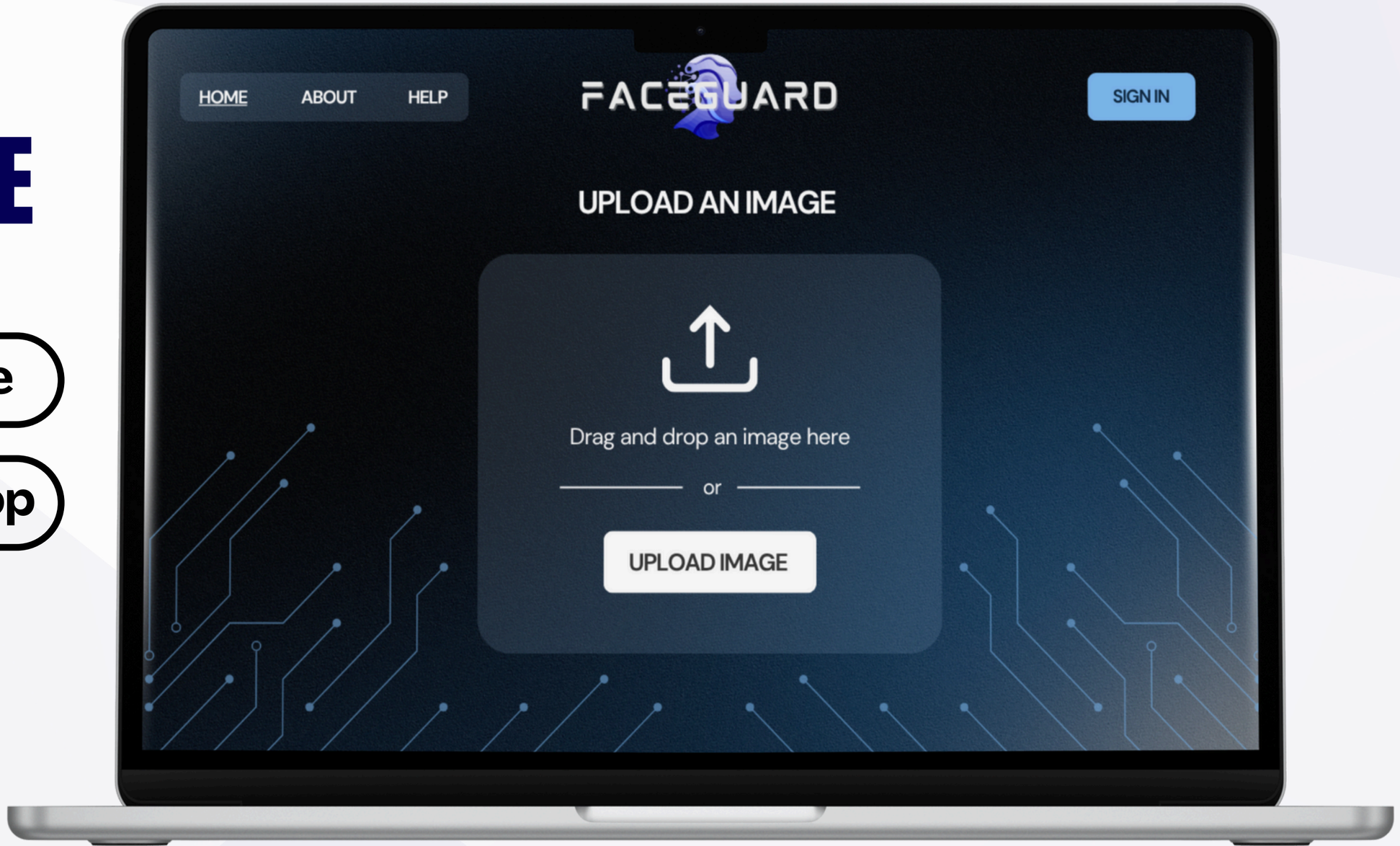
PROJECT PROTOTYPE

- Upload Image
- Drag-and-drop



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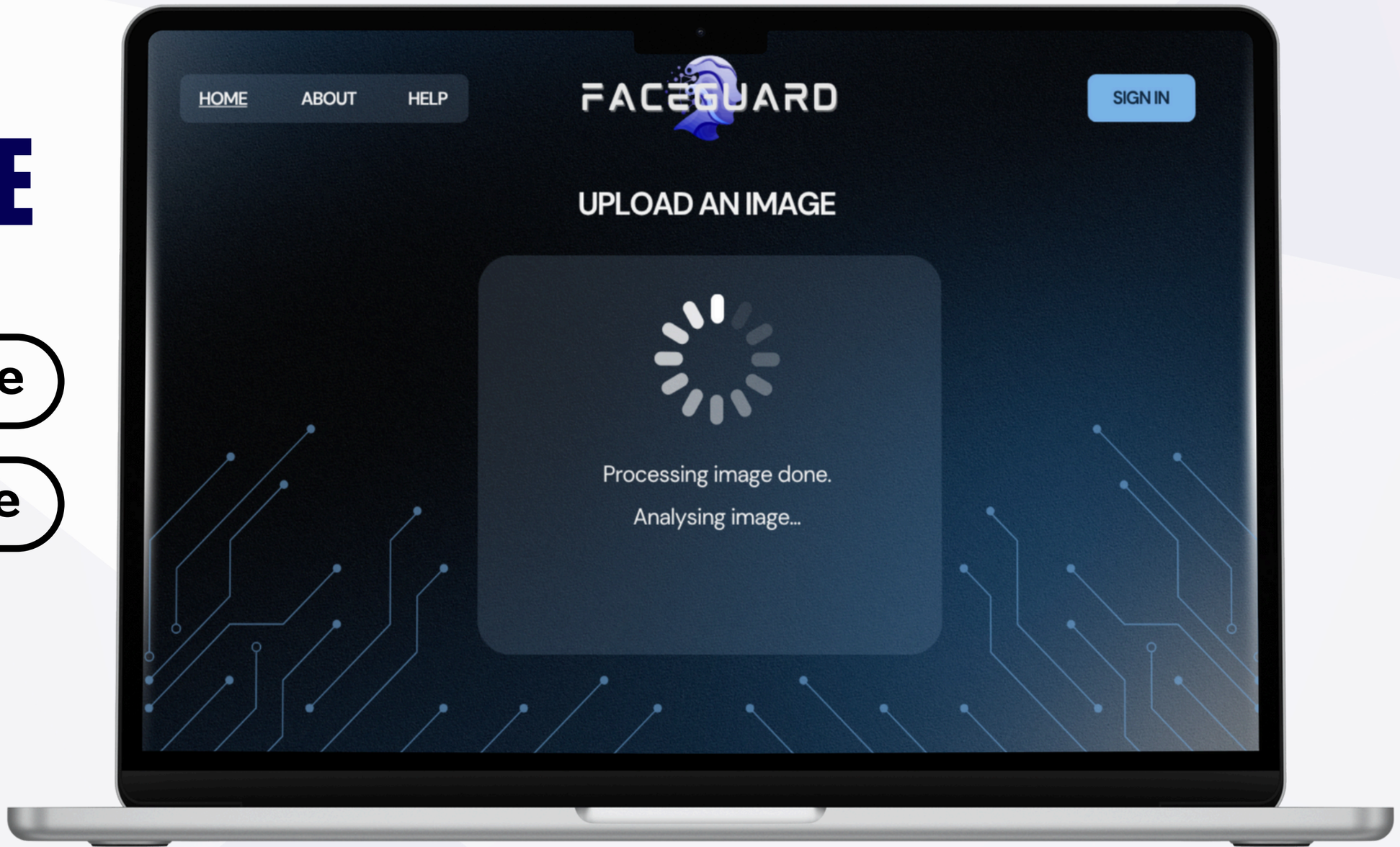
PROJECT PROTOTYPE

- Process Image
- Analyse Image



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PROJECT PROTOTYPE

- Classification Result
- Confidence Score
- Visualisation Explanation

PROJECT FEASIBILITY

Google colab

Cloud-based Training Platform

Google Drive

Project Resource Management

zoom

Project Meeting Platform

Jira

Project Process Management

EXIF

Image Metadata Removal Tool

netlify

Web Hosting Server

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PROJECT FEASIBILITY

PyTorch

Deep Learning Framework

FastAPI

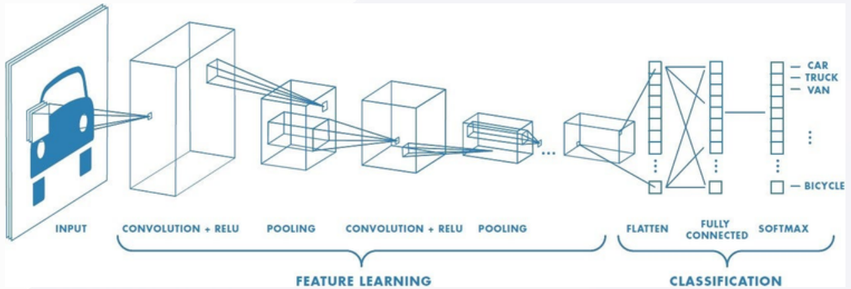
Backend Web Framework

React

Frontend Web Framework

Grad-CAM

Explainability Service

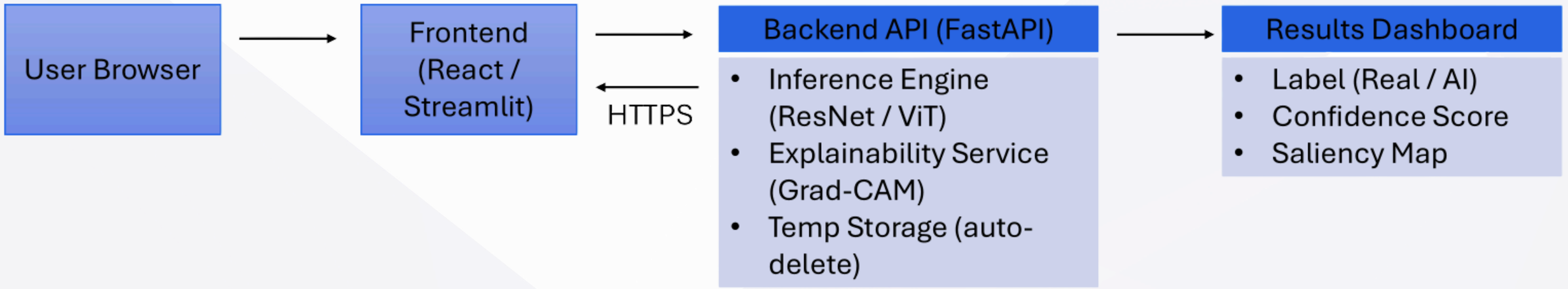


CNN

Inference Engine (ResNet / ViT)

PROJECT FEASIBILITY

System Architecture



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MANAGEMENT

PROJECT MANAGEMENT

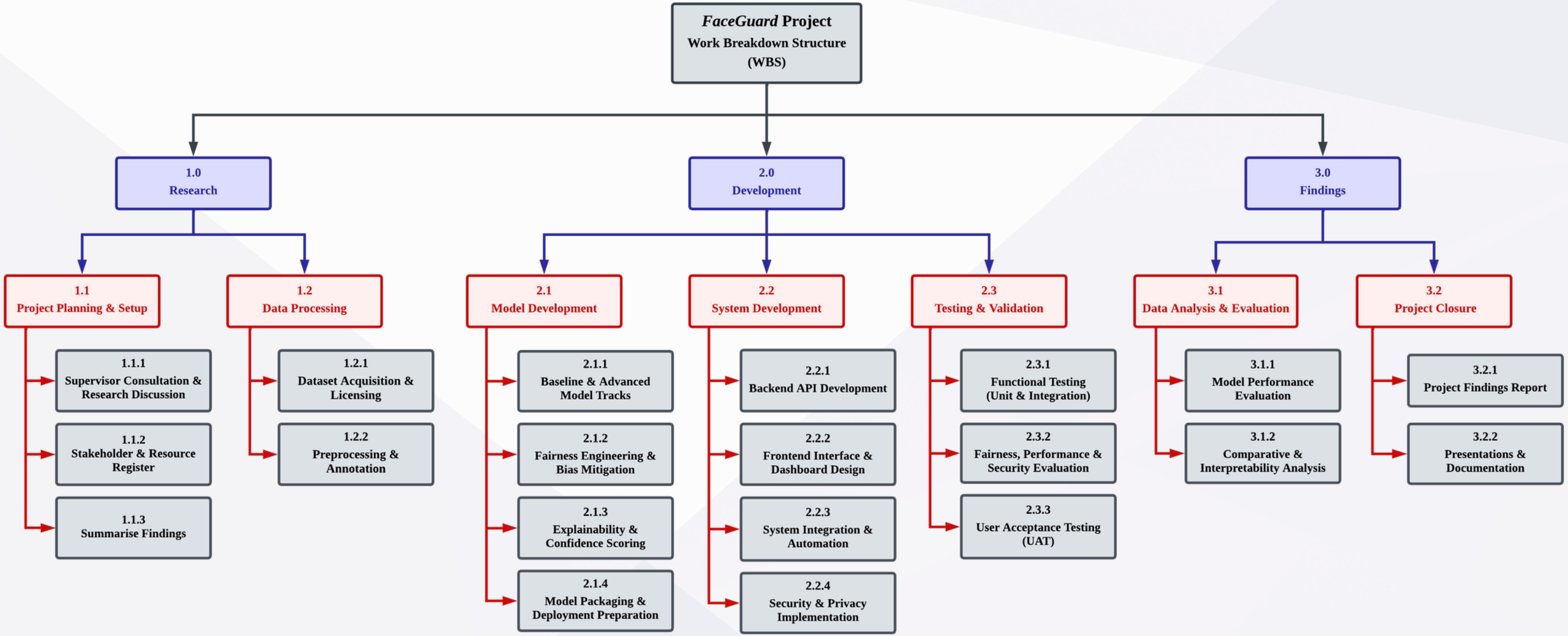
Agile Methodology (Scrum Model)

PROJECT MANAGEMENT

SPRINT GOALS

Sprint 1	Baseline Model Training
Sprint 2	Fairness & Explainability Engineering
Sprint 3	System & Backend Architecturing
Sprint 4	Frontend Features Integration
Sprint 5	Security & Privacy Mechanisms

WORK BREAKDOWN STRUCTURE (WBS)



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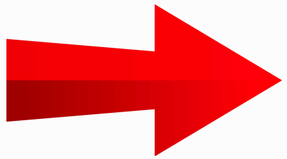
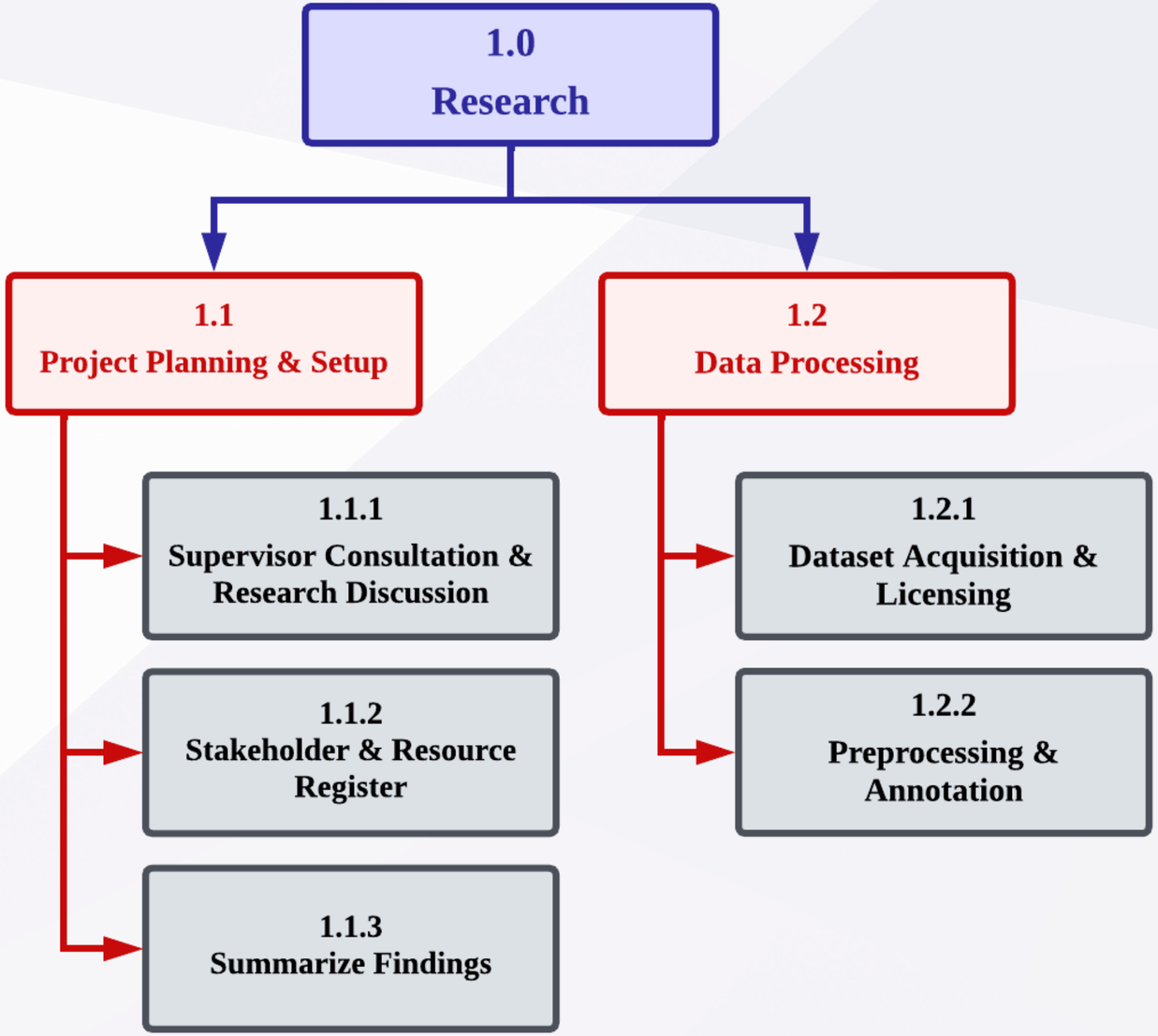
RESEARCH PHASE





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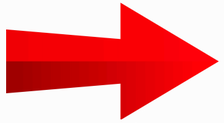
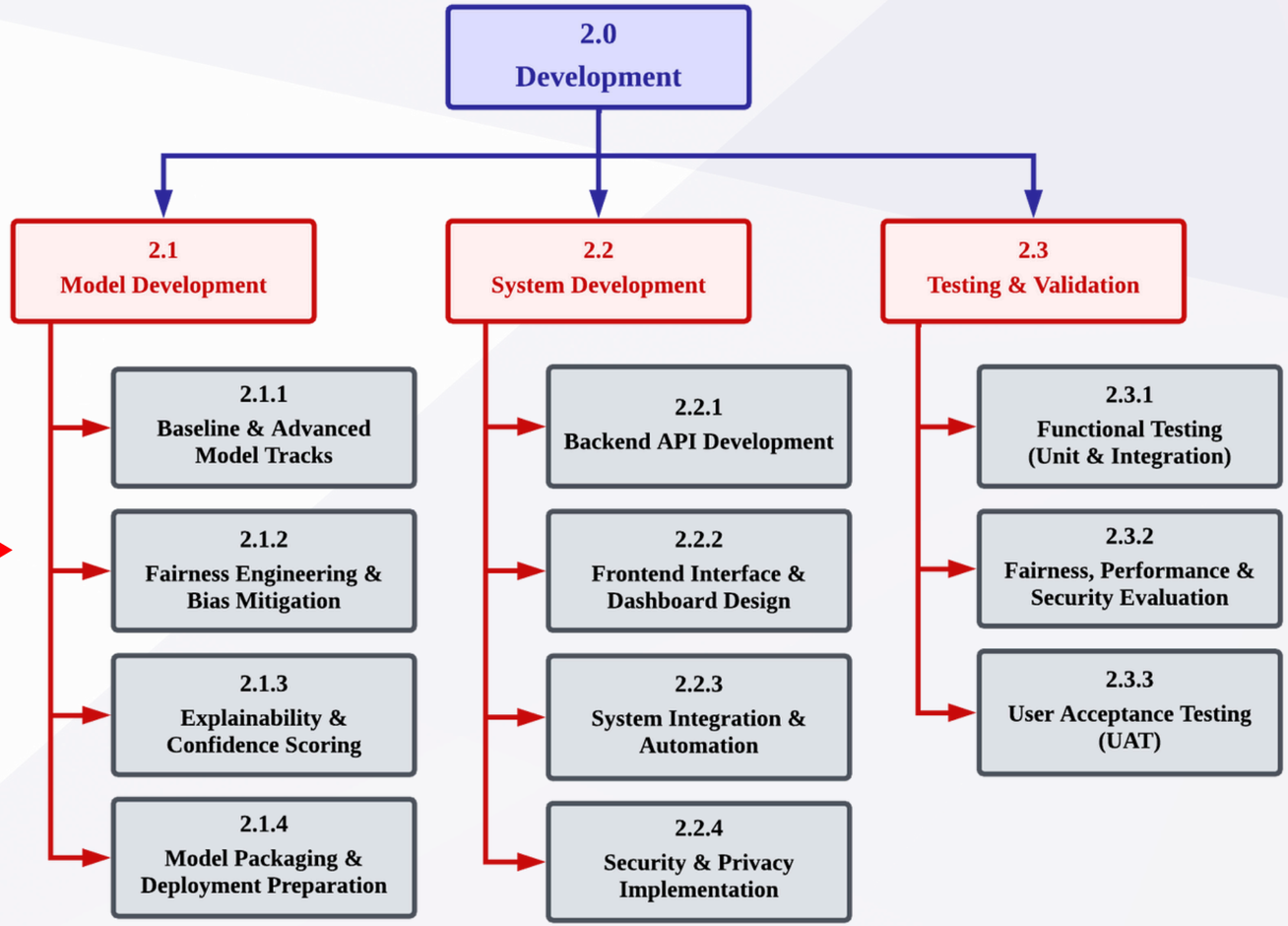
DEVELOPMENT PHASE





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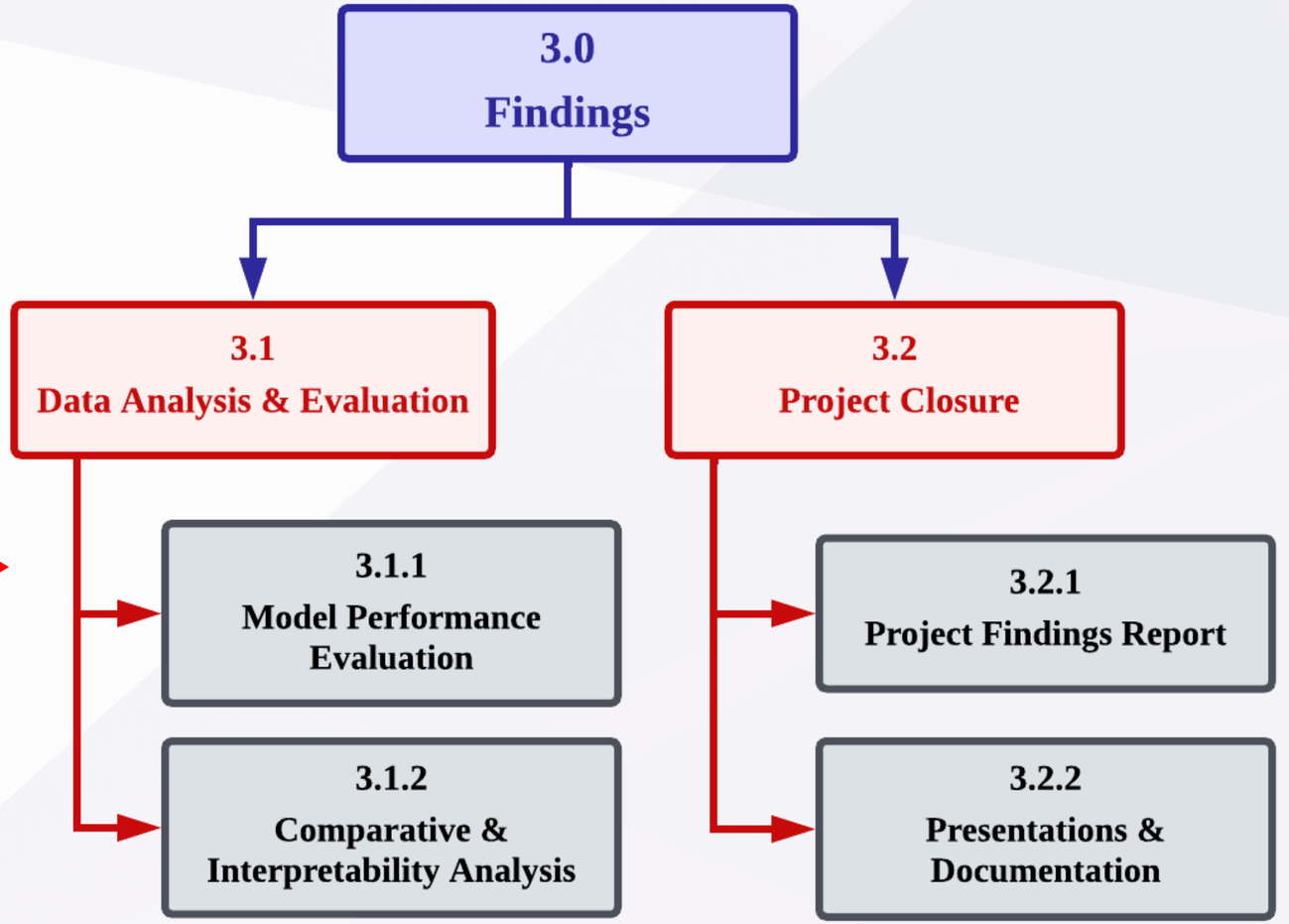
FINDINGS PHASE



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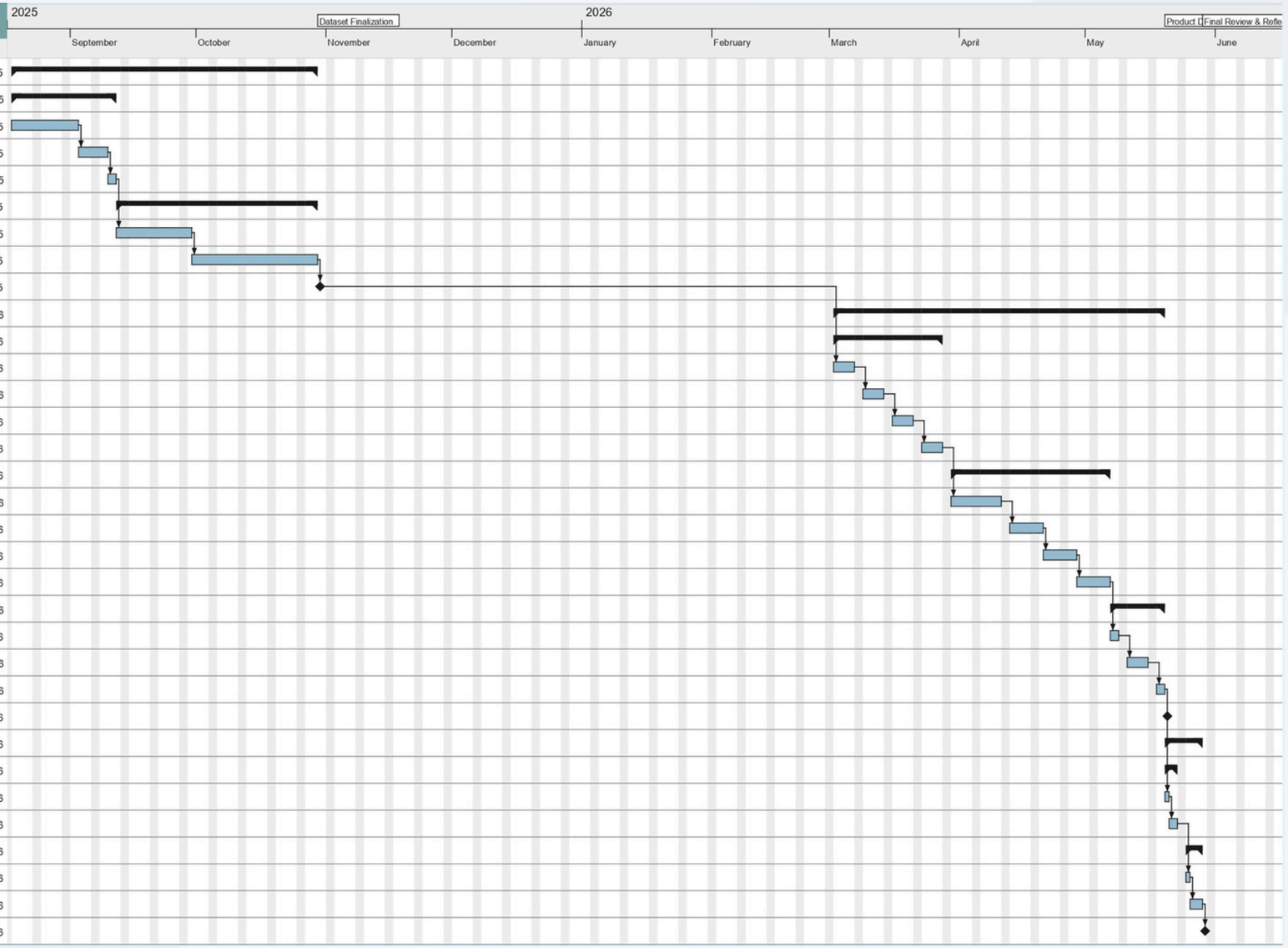
GANTT CHART

Name	Begin date	End date
Research Phase	18/8/25	29/10/25
Project Planning & Setup	18/8/25	11/9/25
Supervisor Consultation and Research Discussion	18/8/25	2/9/25
Stakeholder and Resource Register	3/9/25	9/9/25
Summarize findings	10/9/25	11/9/25
Data Processing	12/9/25	29/10/25
Dataset Acquisition & Licensing	12/9/25	29/9/25
Preprocessing & Annotation	30/9/25	29/10/25
Dataset Finalization	30/10/25	30/10/25
Development Phase	2/3/26	19/5/26
Model Development	2/3/26	27/3/26
Baseline & Advanced Model Tracks	2/3/26	6/3/26
Fairness Engineering & Bias Mitigation	9/3/26	13/3/26
Explainability & Confidence Scoring	16/3/26	20/3/26
Model Packaging & Deployment Preparation	23/3/26	27/3/26
System Development	30/3/26	6/5/26
Backend API Development	30/3/26	10/4/26
Frontend Interface & Dashboard Design	13/4/26	20/4/26
System Integration & Automation	21/4/26	28/4/26
Security & Privacy Implementation	29/4/26	6/5/26
Testing & Validation	7/5/26	19/5/26
Functional Testing (Unit & Integration)	7/5/26	8/5/26
Fairness, Performance & Security Evaluation	11/5/26	15/5/26
User Acceptance Testing (UAT)	18/5/26	19/5/26
Product Deployment	20/5/26	20/5/26
Finding Phase	20/5/26	28/5/26
Data Analysis & Evaluation	20/5/26	22/5/26
Model Performance Evaluation	20/5/26	20/5/26
Comparative & Interpretability Analysis	21/5/26	22/5/26
Project Closure	25/5/26	28/5/26
Prepare Findings Report	25/5/26	25/5/26
Presentations & Documentation	26/5/26	28/5/26
Final Review & Reflection	29/5/26	29/5/26



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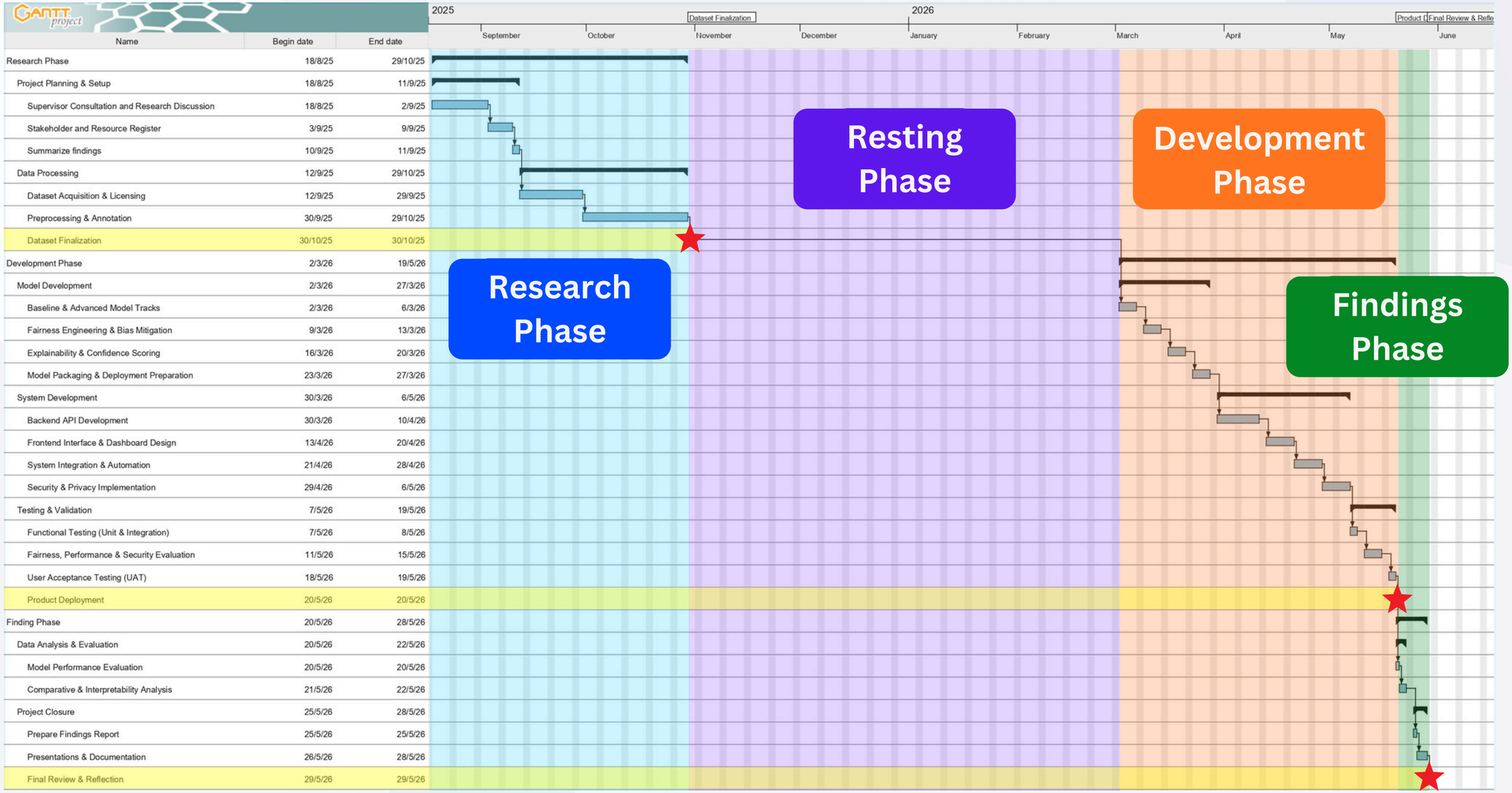

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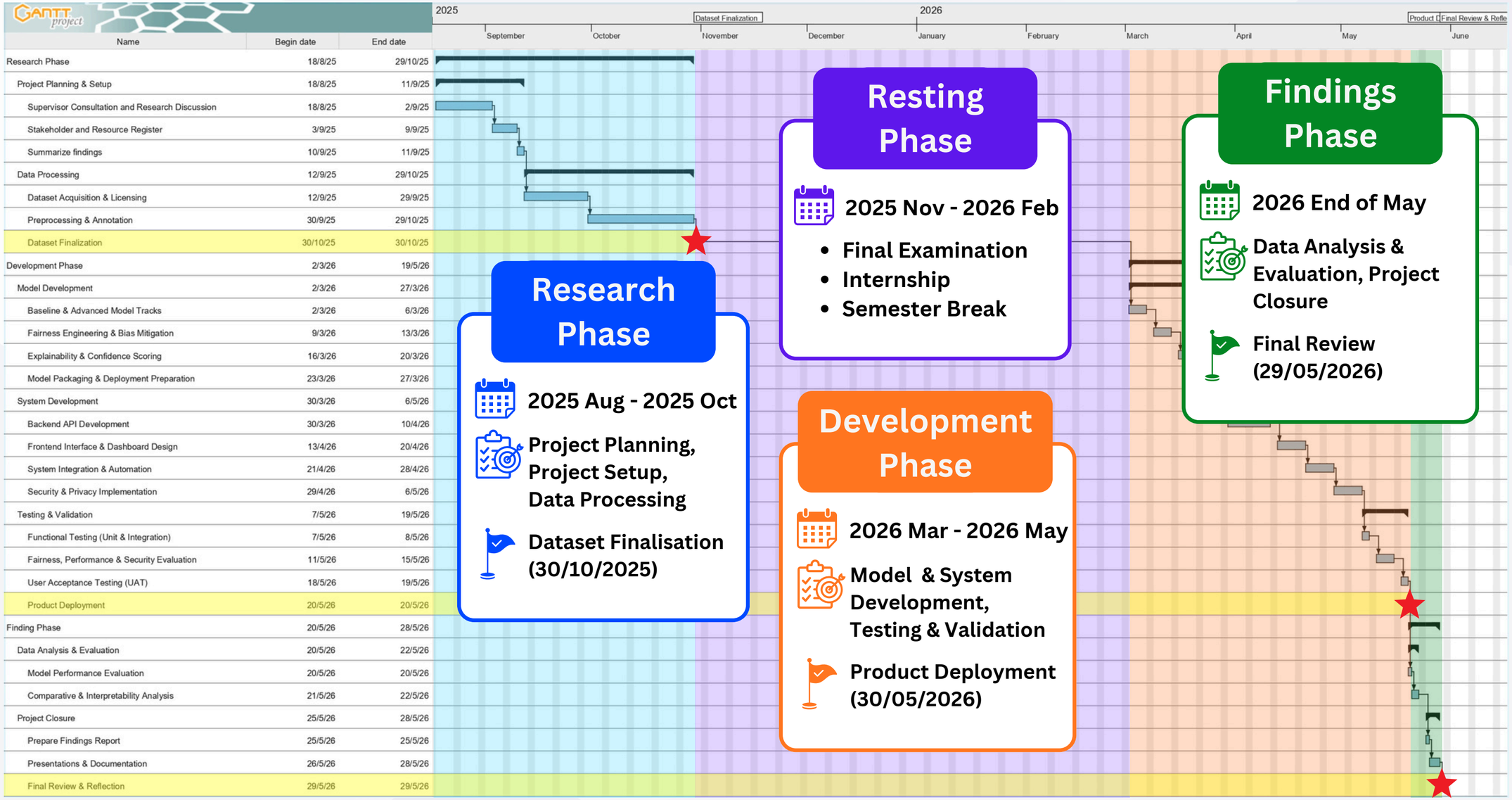

CONCLUSION

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IN-SCOPE

- User-friendly web platform for AI image detection
- Secure image upload with automatic deletion.
- CNN-based binary classifier training with AI-Face v2 data
- Explainability & confidence score feature using Grad-CAM for visual transparency

OUT-OF-SCOPE

- Video deepfake detection
- Mobile app deployment
- Face identification or face matching

PROJECT RISKS

Technical Risk

High

Limited GPU or cloud access may delay model training and experimentation.

Mitigation

Adopt premium cloud credits with Google Colab Pro or access to Monash GPU lab.

Ethical Risk

High

Dataset imbalance across demographic subgroups may introduce bias in results.

Mitigation

Continuously monitor accuracy and precision across key subgroups like skin tone, gender, age

Project Risk

High

Tight project schedules affecting project progress.

Mitigation

Regularprogress reviews and status updates.
Adjust timelines and reallocate resources as needed.

CONCLUSION

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01. Develop a functional AI image detector

02. Using PyTorch and CNN architecture

03. Agile project management methodology

04. 12-months completion by the end of 2026 Semester 1

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